

Py-Java Cheetsheet

py↔Java 1-Liner DSA Initialization Cheat Sheet

Purpose	Python	Java
Empty List	<code>arr = []</code>	<code>ArrayList<Integer> arr = new ArrayList<>();</code>
List with values	<code>arr = [1,2,3]</code>	<code>ArrayList<Integer> arr = new ArrayList<>(Arrays.asList(1,2,3));</code>
Integer Array	<code>arr = [0]*5</code>	<code>int[] arr = new int[5];</code>
2D Array	<code>grid = [[0]*m for _ in range(n)]</code>	<code>int[][] grid = new int[n][m];</code>
String List	<code>arr = ["a","b"]</code>	<code>ArrayList<String> arr = new ArrayList<>(Arrays.asList("a","b"));</code>
Empty Tuple	<code>t = ()</code>	<code>Pair<Integer,Integer> p = new Pair<>(1,2);</code>
Dictionary / HashMap	<code>mp = {}</code>	<code>HashMap<Integer,Integer> mp = new HashMap<>();</code>
String HashMap	<code>mp = {"a":1}</code>	<code>HashMap<String,Integer> mp = new HashMap<>();</code>
Ordered Map	<code>from collections import OrderedDict</code>	<code>TreeMap<Integer,Integer> mp = new TreeMap<>();</code>
Default Dictionary	<code>mp = defaultdict(int)</code>	<code>HashMap<Integer,Integer> mp = new HashMap<>();</code>
Set	<code>s = set()</code>	<code>HashSet<Integer> set = new HashSet<>();</code>
Set with values	<code>s = {1,2,3}</code>	<code>HashSet<Integer> set = new HashSet<>(Arrays.asList(1,2,3));</code>
Sorted Set	<code>s = set(sorted(arr))</code>	<code>TreeSet<Integer> set = new TreeSet<>();</code>
Queue	<code>q = deque()</code>	<code>Queue<Integer> q = new LinkedList<>();</code>
Deque	<code>dq = deque()</code>	<code>Deque<Integer> dq = new ArrayDeque<>();</code>
Stack	<code>st = []</code>	<code>Stack<Integer> st = new Stack<>();</code>
Min Heap	<code>pq = []</code>	<code>PriorityQueue<Integer> pq = new PriorityQueue<>();</code>
Max Heap	<code>pq = []</code>	<code>PriorityQueue<Integer> pq = new PriorityQueue<>(Collections.reverseOrder());</code>
Graph Adjacency List	<code>graph = [[] for _ in range(n)]</code>	<code>ArrayList<Integer>[] graph = new ArrayList[n];</code>
Boolean Array	<code>vis = [False]*n</code>	<code>boolean[] vis = new boolean[n];</code>
Integer Frequency Array	<code>freq = [0]*100</code>	<code>int[] freq = new int[100];</code>
Character Array	<code>chars = list(s)</code>	<code>char[] chars = s.toCharArray();</code>
String Builder	<code>res = []</code>	<code>StringBuilder sb = new StringBuilder();</code>
Infinite Integer	<code>INF = float('inf')</code>	<code>int INF = Integer.MAX_VALUE;</code>
Long Infinite	<code>INF = 10**18</code>	<code>long INF = Long.MAX_VALUE;</code>

Purpose	Python	Java
Mod Constant	<code>MOD = 10**9 + 7</code>	<code>static final int MOD = 1000000007;</code>
Counter	<code>cnt = Counter(arr)</code>	<code>HashMap<Integer,Integer> cnt = new HashMap<>();</code>
Pair List	<code>pairs = [(1,2)]</code>	<code>ArrayList<Pair<Integer,Integer>> pairs = new ArrayList<>();</code>
Matrix List	<code>mat = [[1,2],[3,4]]</code>	<code>int[][] mat = {{1,2},{3,4}};</code>
String Array	<code>arr = ["a","b"]</code>	<code>String[] arr = {"a","b"};</code>
Integer Stream List	<code>arr = list(map(int,input().split()))</code>	<code>int[] arr = Arrays.stream(br.readLine().split("")).mapToInt(Integer::parseInt).toArray();</code>
Clone List	<code>b = a[:]</code>	<code>ArrayList<Integer> b = new ArrayList<>(a);</code>
Copy Array	<code>b = arr.copy()</code>	<code>int[] b = arr.clone();</code>
Reverse List	<code>arr[::-1]</code>	<code>Collections.reverse(arr);</code>
Sort List	<code>arr.sort()</code>	<code>Collections.sort(arr);</code>
Reverse Sort	<code>arr.sort(reverse=True)</code>	<code>arr.sort(Collections.reverseOrder());</code>
Join Strings	<code>" ".join(arr)</code>	<code>String.join(" ", arr);</code>
Split String	<code>s.split()</code>	<code>s.split(" ");</code>
Enumerate	<code>for i,x in enumerate(arr)</code>	<code>for(int i=0;i<arr.size();i++)</code>
Range Loop	<code>for i in range(n)</code>	<code>for(int i=0;i<n;i++)</code>
Lambda Function	<code>lambda x:x*x</code>	<code>x -> x*x</code>
Custom Comparator	<code>key=lambda x:x[0]</code>	<code>(a,b) -> a[0]-b[0]</code>
Read Integer	<code>n = int(input())</code>	<code>int n = sc.nextInt();</code>
Read String	<code>s = input()</code>	<code>String s = sc.next();</code>
Fast Input	<code>sys.stdin.readline</code>	<code>BufferedReader br = new BufferedReader(new InputStreamReader(System.in));</code>
Print	<code>print(x)</code>	<code>System.out.println(x);</code>
Ternary Operator	<code>a if cond else b</code>	<code>cond ? a : b</code>
Membership Check	<code>x in s</code>	<code>set.contains(x)</code>
Append List	<code>arr.append(x)</code>	<code>arr.add(x);</code>
Pop Last	<code>arr.pop()</code>	<code>arr.remove(arr.size()-1);</code>
Queue Push	<code>q.append(x)</code>	<code>q.offer(x);</code>
Queue Pop	<code>q.popleft()</code>	<code>q.poll();</code>
Heap Push	<code>heapq.heappush(pq,x)</code>	<code>pq.offer(x);</code>
Heap Pop	<code>heapq.heappop(pq)</code>	<code>pq.poll();</code>
Heap Top	<code>pq[0]</code>	<code>pq.peek();</code>
Map Insert	<code>mp[k]=v</code>	<code>mp.put(k,v);</code>
Map Get	<code>mp[k]</code>	<code>mp.get(k);</code>
Map Default	<code>mp.get(k,0)</code>	<code>mp.getOrDefault(k,0);</code>
Set Insert	<code>s.add(x)</code>	<code>set.add(x);</code>
Set Remove	<code>s.remove(x)</code>	<code>set.remove(x);</code>
Length	<code>len(arr)</code>	<code>arr.size()</code>
String Length	<code>len(s)</code>	<code>s.length()</code>

Purpose	Python	Java
Max Value	<code>max(arr)</code>	<code>Collections.max(arr);</code>
Min Value	<code>min(arr)</code>	<code>Collections.min(arr);</code>
Sum	<code>sum(arr)</code>	<code>arr.stream().mapToInt(Integer::intValue).sum();</code>
Absolute	<code>abs(x)</code>	<code>Math.abs(x);</code>
Square Root	<code>math.sqrt(x)</code>	<code>Math.sqrt(x);</code>
Power	<code>pow(a,b)</code>	<code>Math.pow(a,b);</code>
Ceiling	<code>math.ceil(x)</code>	<code>Math.ceil(x);</code>
Floor	<code>math.floor(x)</code>	<code>Math.floor(x);</code>

Most Important Imports in Java

```
import java.util.*;
import java.io.*;
```

JAVA

Most Used Java DSA Initializations

```
ArrayList<Integer> arr = new ArrayList<>();

HashMap<Integer,Integer> mp = new HashMap<>();

HashSet<Integer> set = new HashSet<>();

Queue<Integer> q = new LinkedList<>();

Deque<Integer> dq = new ArrayDeque<>();

Stack<Integer> st = new Stack<>();

PriorityQueue<Integer> minHeap = new PriorityQueue<>();

PriorityQueue<Integer> maxHeap =
    new PriorityQueue<>(Collections.reverseOrder());

TreeMap<Integer,Integer> tm = new TreeMap<>();

TreeSet<Integer> ts = new TreeSet<>();
```

JAVA